

Comprehensive Literature Review for Assessing knowledge graph quality: Frameworks and catalogs

The following table present a structured, analytical comparison of **36 key studies**:

S. N	Title	Author(s)	Publish ed year	Abstract	Objectives	Methodolog y	Algorithm used	Top Model	Accurac y	Advantages	Drawback s	Dimensions	Definition of each dimension	Reference
1	Quality assessment for Linked Data: A Survey	Amrapali Zaveri, Anisa Rula Andrea Maurino, Ricardo Pietrobon, Jens Lehmann and Sören Auer.	2015	Summarizes linked data quality assessment papers, introducing existing methods, and categorizing dimensions into six groups.	To provide a theoretical foundation for future research and propose a unified description of data quality terms, dimensions, and measurement indicators.	Literature review and categorization; conceptual proposal for unified terminology.	N/A (Survey/Review)	Foundation al Taxonomy	N/A	Comprehensive foundational categorization (6 dimensions); establishes baseline terminology for the domain.	Did not specify tooling; lack of focus on end-to-end operational systems or large-scale KG-specific algorithms.	Availability, Licensing, Interlinking, Security, Performance, Syntactic Validity, Semantic Accuracy, Consistency, Conciseness, Completeness, Relevancy, Trustworthiness, Understandability, Timeliness, Representational-Conciseness, Interoperability, Interpretability, Versatility.	Availability: The extent to which data (or some portion of it) is present, obtainable and ready for use. Licensing: The granting of permission for a consumer to re-use a dataset under defined conditions. Interlinking: The degree to which entities that represent the same concept are linked to each other, be it within or between two or more data sources. Security: The extent to which data is protected against alteration and misuse. Performance: The efficiency of a system that binds to a large dataset, where a more performant source allows for more efficient data processing. Syntactic Validity: The degree to which an RDF document conforms to the specification of the serialization format. Semantic Accuracy: The degree to which data values correctly represent the real world facts. Consistency: A state where a knowledge base is free of (logical/formal) contradictions with respect to particular knowledge representation and inference mechanisms. Conciseness: The minimization of redundancy of entities at the schema and the data level. Completeness: The degree to which all required information is present in a particular dataset. Relevancy: The provision of information which is in accordance with the task at hand and important to the users’ query. Trustworthiness: The degree to which the information is accepted to be correct, true, real and credible. Understandability: The ease with which data can be comprehended without ambiguity and be used by a human information consumer. Timeliness: A measure of how up-to-date data is relative to a specific task. Representational-Conciseness: A representation of the data which is compact and well formatted on the one hand and clear and complete on the other hand. Interoperability: The degree to which the format and structure of the information conforms to previously returned information as well as data from other sources. Interpretability: Technical aspects of the data regarding whether information is represented using an appropriate notation and whether the machine is able to process the data. Versatility: The availability of the data in different representations and in an internationalized way.	<i>A. Zaveri, A. Rula, A. Maurino, R. Pietrobon, J. Lehmann, and S. Auer, "Quality assessment for Linked Data: A Survey," Semantic Web, vol. 7, no. 1, pp. 63-93, 2015. Read</i>
2	Linked Data Quality Assessment: A Survey	Aparna Nayak, Bojan Bozic, and Luca Longo.	2022	Identifies state of the art, highlighting fragmentation, and proposing a solution based on	To identify existing end-to-end frameworks and propose a standard scalable system that integrates	Survey analysis; proposed an ontology-based solution (FSDCQ ontology) to analyze root causes of	Ontology-based reasoning (SWRL), Fuzzy Inference	FSDCQ Ontology	Equally good as k-NN (for feature selection recommendation)	Highlights the critical need for end-to-end refinement systems; identifies core limitations of prior work.	Most frameworks reviewed were fragmented, dealing with only one aspect of quality; many	Availability, Licensing, Interlinking, Security, Performance, Syntactic Validity, Semantic Accuracy,	Availability: Focuses on the ability to access and retrieve data. The paper highlights tools like LD Sniffer specifically designed to analyze the availability of a given URI. Licensing: Concerns the specific permissions and legal rights required to access or use the data. This is part of the broader effort to analyze the legal aspects of data retrieval for particular use cases. Interlinking: Relates to the connection and retrieval of data across various different sources. This is crucial because integrating data from multiple sources can sometimes lead to contradictory information if the interlinking is not managed	<i>A. Nayak, B. Bozic, and L. Longo, "Linked Data Quality Assessment: A Survey," in Proc. Int. Conf. Information Technology and</i>

				ontology to build an end-to-end system.	quality issue identification, evaluation, and improvement.	quality violations.					focused solely on DBpedia.	Consistency, Conciseness, Completeness, Relevancy, Trustworthiness, Understandability, Timeliness, Representational-Conciseness, Interoperability, Interpretability, Versatility. correctly. Security: Involves the authentication protocols and protected access required for data retrieval. It is one of the key aspects of the "accessibility" group that ensures data is obtained through authorized channels. Performance: Refers to the technical efficiency and speed with which data can be retrieved. The paper notes that many current studies prioritize scalability performance as a primary evaluation method for assessment tools. Syntactic Validity: Ensures data follows specific rules, such as having no malformed datatype literals. For example, the paper describes a violation where a "paper count" is recorded as a decimal (12.3) when the expected datatype is an integer. Semantic Accuracy: Measures whether the information correctly represents the real-world entities it describes. The sources suggest using external sources (like comparing triples to Google Maps) to identify inaccurate information and generate a reliability score. Consistency: Focuses on whether the information is logically coherent and free of internal contradictions. This dimension can be evaluated using a reasoning engine to infer triples that do not obey the logical rules mentioned in a knowledge base. Conciseness: Relates to the logical efficiency and lack of redundancy within the data representation. It is an "intrinsic" dimension, meaning it focuses on whether the data is logically consistent in itself. Completeness: Measures the degree to which all necessary information required to represent the real world is present. Common issues that degrade this dimension include missing data or missing entity relationships. Relevancy: The extent to which the data is applicable and helpful for a consumer's specific task. Assessment of relevancy helps analysts publish better quality data by understanding how it fits the needs of its end-users. Trustworthiness: The degree to which the information is considered reliable and credible within its specific context of use. It is considered a highly context-dependent dimension, as what is trustworthy for one user may not be for another. Understandability (Comprehendibility): The ease with which a user can grasp and interpret the data. The paper suggests that identifying the root causes of errors can help both novice and domain experts better comprehend the quality of the data. Timeliness: Concerns how up-to-date the information is relative to the current time or task requirements. Some assessment tools, like Sieve, improve quality by choosing the most accurate value based on "time-closeness". Representational-Conciseness (Rep-Conciseness): Focuses on how compactly and efficiently the data format and design are structured. This is a "representational" dimension that captures information about the data's design. Interoperability: The ability of data to be read and understood by both humans and machines. This is the main goal of the Semantic Web, and following these principles allows for better data exchange across systems. Interpretability: Relates to how well the design of the data facilitates a correct understanding of its meaning. Tools that provide statistical information (like the number of triples or frequent patterns) help users interpret the underlying quality of the dataset. Versatility: Concerns the flexibility of the data design, allowing it to be used effectively across different applications or formats. It is grouped with other dimensions that focus specifically on the format and technical structure of the data representation.	<i>Applications (ICITA), 2022.</i> Read
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3	Luzzu - A Methodology and Framework for Linked Data Quality Assessment	Jeremy Debbattista, Sören Auer, and Christoph Lange.	2016	Presents Luzzu, a framework for Linked Data Quality Assessment based on an extensible interface and an ontology-driven back-end.	To define a holistic data quality lifecycle and provide a framework that is scalable, extensible, and interoperable for assessing Linked Data quality.	Conceptual methodology; Framework implementation using a scalable stream processor for efficiency, and an ontology-driven back-end for quality metadata.	Stream processing	Luzzu Framework	N/A (Tool/Framework)	Scalable through stream processing; highly extensible via interfaces for new metrics; interoperable via DQV-compatible metadata.	Early ranking algorithms lacked sophisticated tie-breakers based on internal dataset characteristics (e.g., number of triples).	Availability, Performance, Licencing, Interoperability, Provenance, Representational Conciseness, Versatility, Interpretability, Consistency.	Availability: This dimension is part of the Accessibility category. It is evaluated using metrics such as "RDF Availability," "End Point Availability," and "No Misreported Content Types". Performance: Also classified under Accessibility, this dimension measures the efficiency of data access. Metrics include "Scalability of Data Source," "Low Latency," "High Throughput," and "Correct URI Usage". Licencing: This is an Accessibility dimension that checks for the existence of machine-readable and human-readable indications of a license. Interoperability: Classified under the Representational category, this dimension involves metrics like the "re-use of Existing Terms" and "re-use of Existing Vocabularies". Provenance: The paper describes Provenance as a dimension for which the authors defined specific metrics, such as "Basic Provenance" (checking for dc:creator or dc:publisher) and "Extended Provenance" (tracking the origin and history of data). Representational Conciseness: This dimension belongs to the Representational category and addresses data clarity and compactness. Metrics include "No Prolix RDF" and the use of "Short URIs". Versatility: Another Representational dimension, it refers to the availability of data in different formats (serializations) and languages for both human and machine consumption. Interpretability: Part of the Representational category, it checks if a machine can process the data, focusing on metrics like "Low Blank Node Usage" and "Defined Classes and Properties". Consistency: This is an Intrinsic dimension that ensures a knowledge base is free from contradictions. It is measured by checking for misplaced classes/properties, disjoint classes, and the correct usage of OWL datatype or object properties.	<i>J. Debbattista, S. Auer, and C. Lange, "Luzzu—A Methodology and Framework for Linked Data Quality Assessment," Journal of Data and Information Quality, vol. 8, no. 1, 2016.</i> Read
4	Efficient Knowledge Graph Accuracy Evaluation	Junyang Gao, Xian Li, Yifan Ethan Xu, Bunyamin Sisman, Xin Luna Dong, Jun Yang.	2019	Proposes an efficient sampling framework to estimate large-scale KG accuracy with strong statistical guarantees while minimizing the cost of human annotation.	To obtain statistically meaningful estimates for accuracy evaluation while dramatically minimizing the required human efforts (annotation costs).	Application of sampling theory; analysis of annotation cost functions; extension for dynamic/incremental evaluation.	Cluster Sampling, Stratified Sampling, Weighted Reservoir Sampling	Efficient Sampling Framework	Up to 60% cost reduction (static KG); Up to 80% cost reduction (evolving KG)	Makes large-scale, gold-standard accuracy checks economically feasible; provides statistically guaranteed estimates for precision/recall.	Focuses predominantly on the Accuracy dimension, requiring external human annotation expertise.	Accuracy, Cost Effectiveness (Value added), Timeliness (Efficiency focus)	Accuracy: The percentage of triples in the knowledge graph that are correct, meaning the stated relationship is consistent with real-life fact. It provides a strong statistical guarantee of quality by ensuring the estimate is unbiased and adheres to a user-defined Margin of Error (MoE) and confidence level. Cost Effectiveness (Value Added): The minimization of manual effort required to obtain statistically meaningful estimates by optimizing the annotation process. It adds value by utilizing cluster sampling to reduce the cost of "Entity Identification," allowing humans to verify multiple facts about a single entity in one go. Timeliness (Efficiency Focus): The speed and responsiveness of the evaluation system, particularly its ability to provide results for both static and rapidly evolving graphs. This focus ensures rapid monitoring by using an iterative process that stops immediately when quality thresholds are met and reuses previous data to evaluate new graph updates.	<i>J. Gao, X. Li, Y. E. Xu, B. Sisman, X. L. Dong, and J. Yang, "Efficient Knowledge Graph Accuracy Evaluation," Proceedings of the VLDB Endowment (PVLDB), vol. 12, no. 11, pp. 1679-1691, 2019.</i> Read
5	Assessing the Quality of a Knowledge Graph via Link Prediction Tasks (LP-Measure)	Ricky Zhu, Alan Bundy, Fangrong Wang, Xue Li, Kuwabena Nuamah, Lei Xu, Stefano Mauceri, Jeff Z Pan.	2024	Introduces the LP-Measure, an automated approach to assess consistency and redundancy	To automatically assess consistency and redundancy in KGs, serving as an effective, quantitative	Automated evaluation by training link prediction models on the KG and testing their ability to recover intentionally	Link Prediction Models (KG embedding techniques)	LP-Measure	Effective in distinguishing high-quality KGs from low-quality KGs	Automated and scalable; eliminates the need for expensive gold standards or human labor; correlates strongly with	Metric fundamentally assesses consistency/redundancy; cannot assess correctness/completeness directly	Availability, Licensing, Interlinking, Security, Performance, Relevancy, Trustworthiness, Understandability	Semantic Accuracy (Correctness): The degree to which triples correctly reflect real-world facts. The authors note that while this is a highly desirable dimension, it is difficult to measure without "gold standards" or expensive human labor. Completeness: The degree to which the entities and relations of a particular domain are represented in a Knowledge Graph (KG), covering both population and property completeness. The paper finds that their proposed LP-Measure is less effective at distinguishing completeness compared to how well it identifies incorrectness. Consistency: A state where a KG is free of contradictions regarding its	<i>R. Zhu et al., "Assessing the Quality of a Knowledge Graph via Link Prediction Tasks," in Proc. 7th Int. Conf. Natural Language Processing and Information</i>

				using link prediction tasks, eliminating the need for gold standards.	proxy for correctness and completeness.	removed triples.				correctness.	(when gold standards are absent).	ty, Timeliness, Syntactic Validity, Semantic Accuracy (Correctness), Consistency, Conciseness, Completeness, Representational-Conciseness, Interoperability, Interpretability, Versatility.	representation and inference mechanism. This is the primary dimension the LP-Measure assesses, as the authors argue a consistent KG is more likely to be correct. Conciseness: An intrinsic dimension related to the absence of redundant information within the graph. The authors explicitly link this to their tool, stating that the LP-Measure concerns self-consistency and redundancy. Syntactic Validity: The degree to which the data conforms to the formal rules of its representation language. This is viewed as an intrinsic property that, along with consistency, helps determine if the internal logic of the KG is sound. Trustworthiness: A measurement of how much a specific triple in a KG can be trusted. The paper mentions that this can be calculated using neural networks trained to capture triple semantics and global information. Timeliness: Refers to how well a Knowledge Graph evolves over time. This is critical for KGs like Wikidata or DBpedia, which are updated daily or through iterative releases. Relevancy: A dimension concerning whether the data is applicable and helpful for a specific task or domain. It is grouped as a contextual dimension, meaning its value depends heavily on the specific use case of the KG. Understandability: The ease with which the data in the KG can be comprehended by its users. Like relevancy, this is a contextual requirement that focuses on the end-user's experience. Availability: The degree to which the KG data and servers are reachable and accessible. This is categorized as an accessibility dimension and is considered a prerequisite for utilizing any KG. Licensing: Whether a KG has a clear and legal license for its data. The paper identifies this as a dimension specifically important for Linked Data (LD). Interlinking: The degree to which a KG is connected to other external, open knowledge graphs. This is a core goal of Linked Data, which attempts to interconnect all open knowledge sources. Security: Measures and protocols that protect the KG from unauthorized access or data corruption. The authors mark this as a specialized dimension specific to the Linked Data framework. Performance: The speed and efficiency with which a system processes and responds to KG queries. This is another accessibility-focused dimension highlighted as being specific to Linked Data environments. Interpretability: Ensuring that information is presented in a way that is clear and easy for people to grasp. The paper notes that this dimension is designed "for humans" rather than for the automated systems used in their study. Representational-Conciseness: The degree to which the knowledge is structured in a compact and efficient manner. This focuses on the form of the data representation rather than the truthfulness of the content. Interoperability: The ability of the Knowledge Graph to function effectively across different platforms or systems. This dimension ensures the KG is flexible enough to be used in various applications, like recommendation systems or question answering. Versatility: The capacity of the data to be used for multiple purposes or in different formats. This is identified as a representation dimension specific to Linked Data.	<i>Retrieval (NLPIR), pp. 124-129, 2024.</i> Read
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6	KGMM -A Maturity Model for Scholarly Knowledge Graphs based on Intertwined Human-Machine Collaboration	Hassan Hussein, Allard Oelen, Oliver Karras, Sören Auer.	2022	Proposes the Knowledge Graph Maturity Model (KGMM) as a structured framework for evaluating KG maturity across five levels.	To provide a structured approach for assessing maturity, promoting evolutionary curation, and incorporating 20 quality measures.	Development of a graded framework (KGMM) based on the principles of the Capability Maturity Model (CMM), focusing on human-machine collaboration.	N/A (Maturity Model/Framework)	Knowledge Graph Maturity Model (KGMM)	Qualitative (Process Maturity).	Provides a governance layer for continuous improvement; assesses organizational process maturity necessary for sustained quality.	Adoption and long-term industrial impact are still being assessed; theoretical framework.	Accuracy, Completeness, Findability, Accessibility, Interoperability, Reusability, Succinctness.	Accuracy: This is a primary quality dimension that includes measures such as syntactic accuracy, timeliness, correctness, and semantic accuracy. Completeness: This dimension is the focus of Maturity Level 2. It includes measures for trustworthiness, instance completeness, property completeness, and population completeness. Findability: Included as a quality dimension, specifically focusing on identifier stability. It is also noted as one of the key FAIR principles the model aims to implement. Accessibility: This dimension comprises measures like responsiveness, easiness, queryability, and dereferenceability. Like Findability, it is derived from the FAIR principles. Interoperability: This dimension is assessed through provenance, data representation, and trackability. It is also a core component of the FAIR principles cited in the sources. Reusability: This dimension is identified as an "essential" requirement at Maturity Level 3 and includes measures for licenses and general reusability. It is the final component of the FAIR principles mentioned. Succinctness: This dimension is measured by conciseness, which is considered "essential" for making a Knowledge Graph readable and understandable.	<i>H. Hussein, A. Oelen, O. Karras, and S. Auer, "KGMM-A Maturity Model for Scholarly Knowledge Graphs based on Intertwined Human-Machine Collaboration," in Proc. Int. Conf. on Asian Digital Libraries, 2022. Read</i>
7	Structural Quality Metrics to Evaluate Knowledge Graph Quality	Sumin Seo, Heeseon Cheon, Hyunho Kim, Dongseok Hyun.	2022	Presents six structural quality metrics (ICR, IPR, CI, IMI, SPA, SPI) to evaluate KGs based on the specificity and active usage of the underlying ontology (structure).	To devise a measure to compare KG quality based on the premise that structure (=ontology) is a key factor, moving beyond metrics focused solely on size/distribution.	Defined and applied six structural quality metrics to compare six cross-domain KGs (Wikidata, DBpedia, YAGO, Freebase, Google KG, Raftel).	Statistical measures of ontology structure (ICR, IPR, CI, IMI, SPA, SPI)	Structural Metrics Model	N/A (Structural comparison)	Provides a novel viewpoint linking quality directly to schema richness and usage; effective for comparative, high-level assessment.	Focuses on schema quality (T-box); generally omits instance-level semantic correctness; technical part (formal property definition) can be improved.	Consistent representation, Concise representation, Interpretability, Ease of understanding, Structural Consistency.	Consistent representation: This refers to the consistency perspective, which focuses on how logically uniform the data within the knowledge graph is. In practice, researchers achieve this by adjusting hierarchical relationships and refining attributes to remove complexity and ensure a steady, predictable logic throughout the system. Think of this as the "grammar" of the graph—ensuring that the rules of relationships are followed the same way everywhere so that the data never contradicts itself. Concise representation: This dimension evaluates how efficiently and compactly a knowledge graph is organized. The sources describe an ontology as concise when it avoids unnecessary layers of hierarchy; for instance, Freebase is noted for its concise structure because its classes lack parent classes, making the organization very simple and direct. Think of this as a "clutter-free filing system"—it provides all the tools you need without hiding them inside a maze of unnecessary, empty folders. Interpretability: While the paper does not use this as a formal metric name, it addresses the concept through the lens of "explainable AI" and user accessibility. It involves constructing the ontology in a "human readable manner" so that both people and machines can easily grasp the meaning and context of the domains and predicates. Think of this as the "clarity of signage" in a building—it ensures that every label and pointer makes immediate sense to anyone trying to navigate the information. Ease of understanding: This is a data quality perspective that considers how easily a user can interact with the graph, including factors like multilinguality. The paper highlights that when structural complexity increases (such as through many-to-many relationships), the "complexity of understanding and utilization" for the user rises, making the graph harder to use. Think of this as "user-friendliness"—it is the difference between a map written in your own language with clear paths and a tangled web of overlapping roads in a language you don't speak.	<i>S. Seo, H. Cheon, H. Kim, and D. Hyun, "Structural Quality Metrics to Evaluate Knowledge Graphs," arXiv preprint arXiv:2211.10011, 2022. Read</i>

													Structural Consistency: This relates to the paper’s core focus on structural quality metrics, which numerically represent the "internal quality of knowledge graphs". It involves maintaining a granular ontology where classes and properties are refined and used "sufficiently" and "abundantly" to reflect real-world information accurately. Think of this as the "architectural integrity" of a building—ensuring that the skeleton (the ontology) is built with enough detail to support all the rooms (the data) without the whole thing becoming too confusing to navigate.	
8	Introducing the Data Quality Vocabulary (DQV)	Riccardo Albertonia, Antoine Isaacb	2020	Defines the W3C standardized vocabulary for describing quality metadata in a machine-readable format.	To provide a consistent means for potential users to judge a dataset's fitness for purpose by standardizing how quality information (metrics, measurements) is provided.	W3C standardization process; development of an RDF vocabulary, defining concepts like dqv:Metric and dqv:Dimension.	N/A (Vocabulary Definition)	DQV Vocabulary	N/A (Standard)	Ensures interoperability and standardized reporting of quality assessment results across heterogeneous tools and datasets.	DQV is purely a descriptive framework; it does not perform assessment or define a single, formal definition of quality.	Accuracy, Completeness, Consistency, Timeliness, Understandability, Relevancy, Objectivity, Believability, Conciseness.	Completeness: The degree to which all required information is present in a particular dataset. The paper illustrates this dimension through a "population completeness metric," which calculates the ratio between the number of objects represented in the dataset and the number of objects expected to be represented based on the declared scope.	<i>R. Albertoni and A. Isaac, "Introducing the Data Quality Vocabulary (DQV)," Semantic Web, vol. 12, no. 1, pp. 81-97, 2020.</i> Read
9	Data Quality in Context	Diane M. Strong, Yang W. Lee, Richard Y. Wang	1997	Seminal work discussing data quality across various dimensions in organizational contexts.	To shift the focus of data quality discussions beyond mere accuracy to a broader contextual understanding of fitness for use.	Conceptual analysis and development of a multi-dimensional framework for data quality evaluation by data consumers.	N/A (Conceptual Foundations)	Data Quality Dimensions Framework	N/A (Conceptual)	Provided the foundational conceptual basis for subsequent standardized catalogs like ISO 25012.	Did not address the specifics of graph-based data structures or automated assessment algorithms common today.	Accessibility, Accuracy, Appropriate Amount, Believability, Completeness, Concise Representation, Consistent Representation, Ease of Understanding, Interpretability, Objectivity, Relevancy, Reputation, Security, Timeliness, Value Added.	Accessibility: This refers to the ease with which data consumers can retrieve and manipulate data to suit their needs. Even if a system is technically connected, data may be viewed as inaccessible if it requires specialized skills to extract or if the communication lines are unreliable. Accuracy: This is the extent to which data values are correct and match a known standard. Organizations like GoldenAir measure this by comparing computerized inventory system data against physical warehouse counts. Appropriate Amount (referred to as "Amount of data"): This involves maintaining a quantity of data that is sufficient for the task but not overwhelming. The goal is to keep the volume "small enough to be accessible but complete enough to be relevant" to the consumer. Believability: This represents the degree to which data is regarded as true and credible by the user. This dimension is often the first to suffer when a consumer notices "mismatches" or conflicting values between two different data sources. Completeness: This is the degree to which all necessary data is present for a specific task. Operational failures, such as mechanics failing to record part numbers on work forms, are a common cause of incomplete records. Concise Representation: This focuses on whether data is presented in a compact and efficient manner. Providing summarized extracts of massive databases helps consumers focus on relevant details without being slowed by sheer data volume. Consistent Representation: This dimension ensures that data definitions and formats are identical across different systems. Data becomes difficult to integrate when one division stores codes with decimal points while another division stores them without. Ease of Understanding: This refers to the clarity of the data and the user's ability to easily comprehend it. The sources suggest that using technologies like graphical interfaces can significantly improve how easily consumers can use and understand data. Interpretability: This is the consumer’s ability to make sense of the data's specific representation or language. Specialized medical terminology or complex	<i>D. M. Strong, Y. W. Lee, and R. Y. Wang, "Data Quality in Context," Communications of the ACM, vol. 40, no. 5, pp. 103-110, 1997.</i> Read

													coding systems (like DRG codes) can act as a "barrier to accessibility" because they require expert knowledge to decode. Objectivity: This is the extent to which data is free from judgment or subjectivity during the production process. Raw, uninterpreted data is generally viewed as having higher quality than data that has been interpreted or coded by individuals. Relevancy: Data is considered relevant when it is applicable and useful to the specific task at hand. As business environments change—such as a hospital shifting from "fee for service" to "per patient" billing—the data that was previously relevant may no longer be useful for new decision-making tasks. Reputation: This reflects the user’s trust in a data source based on its historical performance or perceived accuracy. A source can develop a poor reputation over time if consumers frequently find inaccuracies, leading them to stop using that data for decisions even if the system is technically "correct". Security (referred to as "Access security"): This involves protecting data confidentiality and managing the permissions required to access it. While necessary for protecting sensitive information like patient records, these security protocols are often perceived by users as a barrier to data accessibility. Timeliness: This refers to data being available and sufficiently up-to-date for the consumer's current needs. <i>Insight from the paper:</i> Processing delays, such as waiting for a weekend batch extract to complete, can lead to timeliness problems that hinder analysis. Value-Added: This dimension measures the extent to which the data provides a clear benefit or advantage to the organization. When data sources are seen as having little value-added due to poor reputation or subjectivity, they are often excluded from the decision-making process.	
10	From Genesis to Maturity: Managing Knowledge Graph Ecosystems Through Life Cycles	Sandra Geisler, Cinzia Cappiello, Irene Celino, David Chaves Fraga, Anastasia Dimou, Ana Iglesias-Molina, Maurizio Lenzerini, Anisa Rula, Dylan Van Assche, Sascha Welten, Maria-Esther Vidal.	2025	Vision paper addressing challenges in managing KGs, proposing KGE ecosystems and life cycles to systematically manage tasks like standardization, continuous updates, and provenance tracking.	To systematically manage KGE tasks to enhance accuracy, consistency, and reliability of KGs.	Vision paper, conceptual framework development focusing on defining KGE life cycles and actors.	N/A (Vision/Conceptual)	Knowledge Graph Ecosystem (KGE) Life Cycle Model	N/A (Conceptual)	Systematically manages creation, maintenance, and evolution; enhances accuracy, consistency, and reliability; addresses scalability of semantic operations.	N/A explicitly mentioned, focuses on identifying and solving critical challenges.	Accuracy, Consistent representation, Timeliness, Reliability, Security, Scalability	Accuracy: This refers to the truthfulness and correctness of data and metadata within the ecosystem. It is maintained through the use of formal constraints (DC) that ensure the quality of all ecosystem components and is a requirement for extracting valuable insights during querying. Consistent Representation (Consistency): This dimension involves the consistent interpretation of diverse data formats using shared vocabularies. It encompasses semantic consistency—ensuring that data integration aligns with the ontology—and logical consistency, which ensures that the ontology, mappings, and the resulting Knowledge Graph do not contradict one another as they evolve. Timeliness (Up-to-date): While the paper primarily uses the term "up-to-date," it defines this as the necessity for a KG to reflect evolving clinical guidelines, records, and research. It is considered a decisive factor for a KG to remain effective and comprehensive over time. Reliability: This is defined as the trustworthiness and robustness of the knowledge graph. It is achieved by formalizing the life cycle—implementing clear processes for updates and integration—and ensures that the ecosystem remains dependable even when handling incremental changes or growing data volumes. Security: This dimension focuses on preserving privacy and data sovereignty. It is defined by the ability of organizations to securely collaborate on data analysis without transferring sensitive information outside of their own systems, often implemented through role-based access controls. Scalability: This is the capacity of the ecosystem to handle large-scale data and computationally expensive reasoning tasks. It involves optimizing performance to manage growing data volumes and user demands without compromising the efficiency or reliability of querying.	<i>S. Geisler et al., "From Genesis to Maturity: Managing Knowledge Graph Ecosystems Through Life Cycles," Proceedings of the VLDB Endowment (PVLDB), vol. 18, no. 5, pp. 1390-1397, 2025. Read</i>
11	Contextual knowledge	Guang-Li Huang,	2024	Proposes a	To identify bias in AI/ML	Contextual KG approach;	N/A (Methodology)	Contextual Knowledge	More effective in	Combines contextual	N/A explicitly mentioned.	Objectivity (Bias),	Fairness: The authors provide a specific definition for this central concept: in the decision-making process, fairness is the "absence of any bias or favouritism based	<i>G. L. Huang and A. Zaslavsky,</i>

	graph approach to bias-reduced decision support systems	Arkady Zaslavsky.		contextual KG approach to capture relationships between task/features/context, identifying bias in AI/ML model datasets. Uses debiased datasets for fairer decision-making.	datasets and support fairer decision-making.	proposed three bias assessment metrics (label, sampling, timeliness bias).	y, leverages ML models)	Graph Approach	supporting fairer decision-making than existing methods (Experimental results)	knowledge for fairer decisions; defines specific bias metrics (label, sampling, timeliness bias).		Timeliness, Accuracy, Relevancy, Value added, Fairness.	on hidden or neglected features against a certain category". It is also categorized into three types: individual, group, and causality-based fairness. Timeliness (Bias): The sources define "Timeliness Bias" as the "change in the context represented before and after modelling". This occurs when a trained model fails to adapt to new data or changing situations, leading to biased predictions in the post-modeling (operational) phase. Accuracy: While not given a philosophical definition, accuracy is treated as a standard performance metric (ACC) and defined as the "proportion of correctly predicted samples to the total number of samples". The paper notes that in existing literature, fairness is often viewed as a "trade-off for the accuracy of the classifiers". Relevancy: A decision support system is defined as a tool designed to provide the "most relevant information of the current context" to support decision-making. The paper's framework focuses on extracting "contextual information relevant to the task" to identify biases. Objectivity (Bias): The sources treat objectivity as a goal of AI, asking if machine learning predictions are truly "fair and objective". Bias itself is defined as a "lack of diversity in a data set" or a situation where a model learns patterns that favor one category over others. Value added: This specific phrase is not mentioned in the sources. While the paper discusses the "contributions" of its research and the "effectiveness" of its methods, it does not define or use "Value added" as a dimension of data or system quality.	<i>"Contextual knowledge graph approach to bias-reduced decision support systems," Journal of Decision Systems, vol. 33, no. suppl, pp. 29-46, 2024.</i> Read
12	Beyond Accuracy: What data quality means to Data Consumers	Richard Y. Wang, Diane M. Strong	1996	Shifted the focus of data quality discussions beyond mere accuracy to a broader contextual understanding of data quality for consumers.	To define data quality from the data consumer's perspective.	Conceptual framework development.	N/A (Conceptual)	Data Quality Dimensions Framework	N/A (Conceptual)	Provided the early conceptual basis for subsequent DQ standards and dimensions.	Pre-dates modern graph structures and automated assessment.	Accessibility, Accuracy, Appropriate Amount, Believability, Completeness, Concise Representation, Consistent Representation, Ease of Understanding, Interpretability, Objectivity, Relevancy, Reputation, Security, Timeliness, Value Added.	Accessibility: It is defined as the extent to which data are available or easily and quickly retrievable. This dimension highlights the consumer's requirement to physically reach the data through the information system. Accuracy: It is the extent to which data are correct, reliable, and certified free of error. While many firms focus solely on this dimension, the sources argue it is only one part of a much broader data quality conceptualization. Appropriate Amount: It is the extent to which the quantity or volume of available data is appropriate. This dimension helps manage potential data overload by ensuring the volume matches the consumer's specific needs. Believability: It is the extent to which data are accepted or regarded as true, real, and credible. Data consumers consider this an intrinsic quality, meaning even correct data lacks quality if the consumer does not trust it. Completeness: It is the extent to which data are of sufficient breadth, depth, and scope for the task at hand. This is a contextual dimension, meaning the amount of detail required depends entirely on the requirements of the consumer's specific job. Concise Representation: It is the extent to which data are compactly represented without being overwhelming. To meet this requirement, data should be brief in its presentation yet complete and to the point. Consistent Representation: It is the extent to which data are always presented in the same format and are compatible with previous data. Consistency ensures that data formats do not change unexpectedly, allowing consumers to easily compare current results with historical data. Ease of Understanding: It is the extent to which data are clear without ambiguity and easily comprehended. This emphasizes the importance of the system presenting data in an intelligible and readable manner. Interpretability: It is the extent to which data are in appropriate language and units and the data definitions are clear. This ensures consumers can correctly	<i>R. Y. Wang and D. M. Strong, "Beyond Accuracy: What Data Quality Means to Data Consumers," Journal of Management Information Systems, vol. 12, no. 4, pp. 5-33, 1996.</i> Read

												interpret data, such as understanding whether financial figures are in dollars, yen, or specific units like millions. Objectivity: It is the extent to which data are unbiased, unprejudiced, and impartial. Maintaining objectivity is a key requirement for data to be considered high-quality in its own right. Relevancy: It is the extent to which data are applicable and helpful for the task at hand. Even accurate data is considered low quality if it is not pertinent to the consumer's decision-making process. Reputation: It is the extent to which data are trusted or highly regarded in terms of their source or content. Consumers often judge the quality of data based on the standing of the department or source that produced it. Security: It is the extent to which access to data can be restricted and hence kept secure. This system-related dimension is vital for protecting proprietary information from being accessed by competitors. Timeliness: It is the extent to which the age of the data is appropriate for the task at hand. Data must be current enough to be fit for use; for example, combat operations require highly current images for accurate strike planning. Value Added: It is the extent to which data are beneficial and provide advantages from their use. High-quality data should explicitly help a user gain a competitive edge or improve the efficiency of their operations.		
13	FAIR Data Maturity Model: specification and guidelines	Edit Herczog, Keith Russell, and Shelley Stall.	2020	Establishes a common set of core assessment criteria and maturity levels for evaluating FAIRness.	Develop a common set of core assessment criteria for FAIRness and help organizations determine if data resources meet required quality thresholds.	The Working Group established indicators and maturity levels.	N/A (Specification/Guideline)	RDA FAIR Data Maturity Model	N/A (Guideline)	Increases coherence and interoperability of existing FAIR assessment frameworks; provides a basis for benchmarking and improvement. ⁷	N/A explicitly mentioned.	Findability, Accessibility, Interoperability, and Reusability.	Findability: This dimension refers to the practices that enable both machines and humans to locate data and metadata. It is achieved by assigning digital resources globally unique and eternally persistent identifiers (PIDs), providing rich metadata for discovery, and ensuring these records are registered or indexed in searchable resources. Accessibility: This dimension focuses on the ability to retrieve data and metadata using their identifiers through standardized communication protocols. It requires that the protocol is open and free where possible, allows for authentication/authorization when necessary, and ensures that metadata remains available even if the data itself is no longer accessible. Interoperability: This dimension requires that data and metadata use a formal, accessible, shared, and broadly applicable language for knowledge representation. It involves using FAIR-compliant vocabularies and including qualified references to other metadata and data to allow for seamless integration into different workflows and systems. Reusability: This dimension aims to ensure that data is described with a plurality of accurate and relevant attributes so it can be used for different purposes or in new contexts. It requires clear and accessible usage licenses, detailed provenance (origin and history), and compliance with domain-relevant community standards.	<i>E. Herczog, K. Russell, and S. Stall, "FAIR Data Maturity Model: specification and guidelines," RDA FAIR Data Maturity Model Working Group, 2020.</i> Read
14	KGHeartBeat: a Knowledge Graph Quality Assessment Tool.	Maria Angela Pellegrino, Anisa Rula, and Gabriele Tuozzo.	2024	An open-source tool for periodically evaluating the quality of KGs.	To provide continuous, operational quality monitoring.	Open source tool development focusing on periodic assessment.	Quality checks/metrics implemented within the tool	KGHeartBeat.	N/A (Tool/Framework)	Focuses on operational/continuous monitoring; used in comparative studies.	N/A explicitly mentioned.	Accessibility, Timeliness, Completeness, Accuracy, Consistency, Availability.	Accessibility: This is identified as one of the heterogeneous and multiple quality dimensions that the tool assesses. In the context of the Semantic Web, this measures the ease with which data can be retrieved by both human users and automated crawlers. Timeliness: Defined specifically as the presence or frequency of data validation. It is considered a crucial dimension for use cases like debunking misinformation and automatic fact-checking. This dimension evaluates how "fresh" the data is and whether the Knowledge Graph (KG) is updated frequently enough to remain relevant. Completeness: This is listed as a fundamental quality dimension within the framework. While not explicitly defined by a single sentence, the paper links it to the "Amount of data," which tracks the number of triples, properties, and entities.	<i>M. A. Pellegrino, A. Rula, and G. Tuozzo, "KGHeartBeat: A Knowledge Graph Quality Assessment Tool," in The Semantic Web: ESWC 2024 Satellite Events, pp. 154-158, 2024.</i> Read

													Semantic Accuracy: The tool includes this as a selectable quality dimension in its user interface. This measures how accurately the digital data reflects real-world truths and whether the meanings assigned to data points are correct. Availability: This dimension measures the operational status of the KG through metrics like SPARQL endpoint availability, the presence of RDF dumps, the dereferenceability of URIs, and the check for inactive links. Availability is the "heartbeat" of the data; if the SPARQL endpoint is offline, the overall quality score of the KG drops significantly because the data cannot be reached. Consistency: This dimension evaluates structural integrity by checking for the use of deprecated classes or properties, invalid usage of undefined classes/properties, and misplaced classes/properties. This ensures the data follows its own established rules and logic, preventing "garbage" entries that don't fit the specified schema.	
15	Quality Without Borders: A Modular Approach to Unified Knowledge Graph Assessment.	Gabriele Tuozzo.	2024/25	Proposes a comprehensive Shared Framework to formally align KG-specific quality metrics, FAIR principles, and the 5-star open data scheme.	Design a Shared Framework to systematically align existing quality paradigms and improve KG Findability.	Systematic analysis and alignment; modular KG aggregator using crawlers, search APIs, and LLMs.	LLM-guided discovery for aggregation	Shared Framework / "KG Weather Station"	N/A (Conceptual/Proposal)	Unified assessment, metric reusability, consistent/comparable results, enhanced understanding, improved Findability.	Preliminary mapping is incomplete; certain FAIR principles (R1.3, A2) require new metrics; lack of validation across diverse domains.	Accessibility, Findability.	Accessibility: The degree to which data can be retrieved by users or machines using standard communication protocols (This definition is not from the sources). The sources map this dimension to the presence of "Working access point(s)" and note that it is often compromised by inactive endpoints or static data dumps that lack live querying functionality. Findability: The ease with which a dataset can be discovered through unique, persistent identifiers and metadata indexing (This definition is not from the sources). The sources identify this as a critical challenge for current Knowledge Graphs due to "link decay" and the fact that major catalogs like the LOD Cloud are not consistently maintained.	<i>G. Tuozzo, "Quality Without Borders: A Modular Approach to Unified Knowledge Graph Assessment," in ISWC 2025 Companion Volume, 2025.</i> Read
16	ISO/IEC 25012 Data Quality Model.	International Organization for Standardization (ISO), International Electrotechnical Commission (IEC), Joint Technical Committee ISO/IEC JTC 1, Subcommittee SC 7 (Software and systems engineering)	2014	Defines 15 data quality characteristics categorized by Inherent (data itself) and System-Dependent (data quality preserved in a computer system) viewpoints.	To provide standardized definitions for data quality characteristics.	Standardization process, conceptual definition of characteristics.	N/A (Standard)	ISO/IEC 25012 Standard	N/A (Standard)	Provides comprehensive, standardized vocabulary for DQ characteristics (15 dimensions).	Generic to all data, not specific to graph structures or semantic relationships.	Accuracy, Completeness, Consistency, Credibility, Currentness, Accessibility, Compliance, Confidentiality, Efficiency, Precision, Traceability, Understandability, Availability, Portability, Recoverability.	Accuracy: The degree to which data attributes correctly represent the true value of the intended object. This is like a GPS coordinate; if the numbers are wrong, you are at the wrong house. Completeness: The degree to which data has values for all expected attributes and related entities. This is like a form where every required field is filled in, leaving no "unknown" gaps. Consistency: The degree to which data is free from contradiction and coherent with other data. If a customer's birthdate says they are 20 years old in one file but 50 in another, the data lacks consistency. Credibility: The degree to which data is regarded as true and believable by users. This is about the "trustworthiness" of the source; you would trust a bank's record of your balance more than a handwritten note. Currentness: The degree to which data is of the right age for its specific context. A weather report from last week is accurate for that day, but it lacks currentness for today's picnic. Accessibility: The degree to which data can be accessed by people who need it in a specific context. If a report is written in a language the user doesn't speak, or is in a locked file, it is not accessible. Compliance: The degree to which data adheres to standards, conventions, or regulations. This is like a tax form that must follow specific legal rules to be accepted by the government. Confidentiality: The degree to which data is protected so only authorized people can see it. This is like a "Digital Vault" that keeps your private medical records away from strangers. Efficiency: The degree to which data can be processed to provide the required performance while using the right amount of resources. This is about how "lean"	<i>ISO/IEC 25012:2008(E), Software engineering — Software product Quality Requirements and Evaluation (SQuaRE) — Data quality model, First edition, 2008-12-15.</i> Read

													the data is; it shouldn't take a supercomputer an hour just to find a single phone number. Precision: The degree to which data attributes are exact or show a high level of detail. Saying it is "3:00 PM" is accurate, but saying it is "3:00:04 PM" provides higher precision. Traceability: The degree to which the history or origin of the data can be tracked. This is like a "Paper Trail" that shows exactly who changed a price in a store's database and when. Understandability: The degree to which data is clear and can be easily interpreted by the user. Using clear labels like "Total Price" instead of "TP_01_final" makes the data much easier to grasp. Availability: The degree to which data can be retrieved by authorized users when they need it. If the server is down, your data might be perfect, but it is "unavailable" to you. Portability: The degree to which data can be moved from one system to another while keeping its quality. This is like a "Universal Plug" that allows your data to work in a new software program without breaking. Recoverability: The degree to which data quality can be maintained or restored even after a system failure. This is like having a "Safety Net" or backup that lets you get your work back if your computer crashes.	
17	From Data Quality to Big Data Quality: A Systematization.	Carlo Batini, Anisa Rula, Monica Scannapieco, and Gianluigi Viscusi.	2015	Examines the relationship between Data Quality and Big Data characteristics (Variety, Volume), focusing on various data types (LOD, sensor data).	Systematize the correlation between structural characteristics and quality dimensions in the context of Big Data.	Systematic literature review and conceptual systematization.	N/A (Conceptual)	Big Data Quality Systematization	N/A (Conceptual)	Provides foundational context for applying DQ principles to Big Data and Linked Open Data.	N/A explicitly mentioned.	Consistency, Accuracy, Completeness, Timeliness, Volume (Appropriate Amount), Variety.	Consistency: This is one of the seven core dimension clusters identified in the sources. It is defined alongside cohesion and coherence as the capability of data to comply with integrity constraints and business rules without contradiction. The paper specifically explores its evolution in the context of sensors and sensor networks, where it includes sub-dimensions like absolute numerical error, hop consistency, and temporal consistency. Accuracy: This is also a primary dimension cluster. It focuses on the adherence of data to the "reality of interest" and includes sub-dimensions such as correctness, validity, and precision. The paper examines its evolution through the lens of maps, detailing types like positional, thematic, and temporal accuracy. Completeness: This dimension belongs to a cluster that includes pertinence and relevance. It refers to representing all relevant aspects of reality. The sources discuss its application in official statistics, focusing on sub-dimensions like representativeness, selectivity, and sparsity. Timeliness: While not its own primary cluster, "Timeliness" is listed as a dimension under the Accuracy cluster for relational data. It is also identified as a necessary property for open data and is discussed in the context of sensor networks as the requirement that data be delivered to the "sink" before or by the time it is expected. Variety: The paper does not categorize "Variety" as a data quality dimension. Instead, it is treated as a core coordinate or structural characteristic of the Big Data phenomenon itself. The authors investigate how the variety of data types (maps, semi-structured text, linked open data) influences the evolution of the other quality dimensions. Volume (Appropriate Amount): Similar to variety, the sources describe "Volume" as a defining characteristic of Big Data (noting it is "too big" for conventional systems), rather than a quality dimension. The closest quality-related concept mentioned is the Redundancy cluster, which includes minimality, compactness, and conciseness, referring to the use of minimal informative resources.	<i>C. Batini, A. Rula, M. Scannapieco, and G. Viscusi, "From Data Quality to Big Data Quality: A Systematization," Journal of Database Management, vol. 26, no. 1, pp. 60-82, 2015.</i> Read
18	A Scalable Framework for	Gezim Sejdiu, Anisa Rula,	2020	Presents an open source	Provide a distributed,	Distributed implementatio	Apache Spark	DistQuality Assessment	N/A (Tool/Fram	First distributed, in-memory	N/A explicitly mentioned.	Timeliness (Performance),	Accuracy: Accuracy is categorized as a core dimension of data quality. The sources note that while many existing solutions focus on a narrow	<i>G. Sejdiu, A. Rula, J. Lehmann, and</i>

	Quality Assessment of RDF Datasets.	Jens Lehmann, and Hajira Jabeen.		implementation of quality assessment of large RDF datasets that can scale out to a cluster of machines.	in-memory approach for computing quality metrics for large RDF datasets.	n using Apache Spark.	(Distributed Computing)	Framework	ework)	approach for quality assessment; scalable to a cluster of machines.		Completeness, Accuracy, Accessibility, Scalability.	aspect—specifically the "syntactic accuracy of literal values"—a comprehensive assessment of accuracy involves capturing various aspects of the underlying data. Completeness: The paper identifies completeness as a fundamental dimension for measuring the quality of Linked Data. It is treated as a heuristic-based measure within the authors' framework to determine how much of the required information is present within a dataset. Timeliness: Timeliness is defined as a dimension along which data quality is measured using specific quality metrics (m1,m2...mk). It reflects how up-to-date or current the RDF data is for its intended applications, such as semantic search or query answering. Accessibility: This dimension refers to the availability and reachable nature of Linked Data resources. The paper highlights that while some previous tools have been designed specifically to assess the "accessibility of resources," the proposed framework is intended to handle accessibility as part of a more generic, large-scale assessment. Performance: In the context of data quality, performance is listed as a dimension used to measure the "fitness for use" of Linked Data. However, the paper more frequently uses "performance" to describe the efficiency of the assessment framework itself, measuring it through "runtime," "complexity," and "speedup". Scalability: While not a dimension of the data itself, scalability is the defining characteristic of the proposed framework. The authors define it through two primary lenses: Size-up Scalability: The ability of the system to process increasingly large volumes of RDF datasets (e.g., scaling from 2GB to 200GB), Node Scalability (Speedup): The ability to reduce execution time linearly by adding more worker nodes to a distributed cluster.	<i>H. Jabeen, "A Scalable Framework for Quality Assessment of RDF Datasets," in The Semantic Web – ISWC 2019: 18th International Semantic Web Conference, Auckland, New Zealand, October 26–30, 2019, Proceedings, Part II, Berlin, Heidelberg: Springer-Verlag, 2019, pp. 261-276. Read</i>
19	A Novel Customizing Knowledge Graph Evaluation Method for Incorporating User Needs.	Ying Zhang and Gang Xiao.	2024	Introduces an accuracy-focused KG evaluation method that incorporates user requirements. It designs an effective two-stage weighted cluster sampling (EP-TWCS) to focus on user-important entities. Experiments show the sampled accuracy closely matches true accuracy with minimal sample size.	Ensure accuracy assessment of KGs in a cost-saving way that meets specific user requirements.	EP-TWCS sampling: assigns weights based on user usage frequency and entity popularity, then performs stratified sampling to estimate KG accuracy.	EP-TWCS (Entity Popularity – Two-stage Weighted Cluster Sampling)	Sampling-based evaluation	Accuracy of sampled assessment nearly equals real accuracy; sample size is minimized.	User-centric: tailors sampling to what users care about; efficient (minimal sample size) while maintaining accuracy.	Focuses only on the accuracy dimension of quality; requires knowledge of user usage weights; may not address completeness or consistency.	Accuracy, Relevancy (User Needs), Value added, Timeliness (Efficiency).	Accuracy: This dimension reflects the degree of correctness of the triples (the fundamental data units) within a knowledge graph. The paper distinguishes between "intrinsic accuracy," which treats all data equally, and "user-need-based accuracy," which focuses on the correctness of entities that are frequently queried and accessed by the user. Relevancy (User Needs): This is defined through the concept of "Fitness for Use," which means the knowledge graph must meet the requirements of specific application scenarios. Relevancy is determined by analyzing user search behavior and search records to ensure the evaluation focuses on the information most relevant to the user’s actual demands. Value Added: This refers to the practical guidance and reference value that the evaluation results provide to the user. The sources argue that for an evaluation to have value, it must be instructive, helping users effectively select the most applicable knowledge graph for their specific tasks rather than just providing a generic accuracy score. Timeliness (Efficiency): This dimension addresses the problem of outdated knowledge and "untimely updating," which can negatively impact a graph's quality. From a process perspective, it also refers to the ability to obtain time-efficient evaluation results in a cost-saving way, reducing the labor and time typically required to assess large-scale graphs.	<i>Y. Zhang and G. Xiao, "A novel customizing knowledge graph evaluation method for incorporating user needs," Scientific Reports, vol. 14, 2024, Art. no. 10.1038/s41598-024-60004-x. Read</i>

20	Continuous Knowledge Graph Quality Assessment through Comparison using ABECTO.	Jan Martin Keil	2024	Presents ABECTO, a command-line tool for continuous QA of RDF knowledge graphs. It automatically compares multiple overlapping KGs to spot missing or incorrect values. The tool outputs quality annotations (deviations, completeness) over time, enabling CI-style monitoring.	Automate periodic monitoring of KG accuracy and completeness without an ideal reference dataset.	CLI framework that takes multiple KGs as input, computes differences (completeness and value deviations) by aligning overlapping entities and relations.	ABECTO (automated graph comparison).	Comparative analysis tool.	Not applicable (qualitative tool); shown to improve detection of omissions and deviations in KG maintenance.	Automates quality checks in CI pipelines; uses existing KGs as “reference” (silver standard); improves detected accuracy/completeness.	Depends on availability of overlapping KGs; may miss defects if all compared KGs share the same errors.	Completeness, Accuracy, Consistent representation, Timeliness, Reproducibility.	Accuracy: The paper defines the assessment of accuracy as the process of monitoring and detecting value deviations. In the context of the ABECTO tool, accuracy is measured by comparing corresponding resources across different knowledge graphs to identify where equivalent variables have non-equivalent values. The sources note that while constraint-based methods (like SHACL) can find implausible values, accuracy assessment through comparison can detect values that are plausible but still incorrect. Completeness: Completeness is defined through the identification of missing data at two levels: - Population Completeness: The detection of resource omissions, which occur when a resource exists in one knowledge graph but has no corresponding resource in another. - Property Value Completeness: The detection of value omissions, where a resource is present in both graphs, but a specific variable has a value in one graph that is missing in the other. The paper aims to measure the "count, coverage, and completeness" of these resources and values to determine the maturity of the knowledge graph. Consistent Representation (Related Concept): The paper addresses the alignment of data structures during the "transformation phase". This phase allows for the "alignment of literal datatypes" and the "deduction of implicit encoded data" to ensure that data from different sources can be compared effectively. Timeliness (Related Concept): The sources do not use the term "Timeliness," but they emphasize the need for "regular" and "continuous" quality assessment. The author argues that because knowledge graphs evolve, quality checks must be integrated into a Continuous Integration (CI) scenario to ensure that errors are caught immediately and that data quality does not decrease over time as the graph grows. Reproducibility (Related Concept): It supports the concept by providing open-source access to the ABECTO tool on GitHub and Zenodo. Furthermore, the author includes "preparation instructions" and "example comparison projects" using real-world data (like Wikidata and DBpedia) so that others can replicate the assessment results.	<i>J. M. Keil, "Continuous Knowledge Graph Quality Assessment through Comparison using ABECTO," in Joint Proceedings of Posters, Demos, Workshops, and Tutorials of the 20th International Conference on Semantic Systems (SEMANTICS-PD WT 2024), Amsterdam, Netherlands, 2024, vol. 3759.</i> Read
21	A Practical Framework for Evaluating the Quality of Knowledge Graphs.	Haihua Chen, Gaohui Cao, Jiangping Chen, and Junhua Ding.	2019	Surveys existing KG evaluation practices and proposes a practical quality framework. It identifies key dimensions (“fit for purpose”), selects metrics per use-case requirements, and suggests evaluation procedures (e.g., scalable sampling for metrics).	Provide a “fit-for-purpose” quality assessment framework by linking user requirements to specific metrics and scalable evaluation methods.	Literature analysis of prior studies; development of a framework mapping use-case needs to choose quality dimensions and metrics, and recommending feasible evaluation approaches (trade-offs between feasibility and scalability).		Conceptual framework		Focuses on practical applicability and scalability of KG QA; covers multiple dimensions guided by requirements.	Conceptual (no empirical validation); framework-level only; actual tool or data not provided.	Relevancy, Accuracy, Completeness, Timeliness.	Relevancy: This is identified as a Contextual dimension. The sources note that for domain-specific tasks, it is an important factor to ensure the knowledge graph is related to that field. Accuracy: This is discussed primarily as semantic accuracy under the Intrinsic category. It is considered a critical requirement for building effective knowledge-based systems and for tasks like data mining, where the accuracy of entity and relation extraction is a precondition. Completeness: This is listed as an Intrinsic quality dimension. The sources mention that if a constructed knowledge graph suffers from incompleteness, the performance of applications like data mining will be affected. Timeliness: This is categorized as a Contextual dimension. It is highlighted as an important factor for enhancing performance in applications such as prediction.	<i>H. Chen, G. Cao, J. Chen, and J. Ding, "A Practical Framework for Evaluating the Quality of Knowledge Graph," in Information Studies and Informatics, 2019, doi: 10.1007/978-981-15-1956-7-10.</i> Read

22	Knowledge Graph Quality Management: A Comprehensive Survey.	Bingcong Xue and Lei Zou.	2023	Provides a systematic survey of knowledge graph quality management, covering everything from theory to practice. It reviews quality issues, dimensions (accuracy, completeness, etc.), metrics, assessment methods, and quality improvement processes.	Summarize and classify research on all aspects of KG quality (issues, dimensions, metrics, tools, processes).	Comprehensive literature review; taxonomy of quality concepts; organization of methods for assessment, detection and completion of KGs; discussion of future directions.		Survey (no single model)		Very comprehensive coverage of KG quality topics; connects theory and practice; identifies open research directions.	Only a survey (no new technique); rapidly becomes outdated as new methods emerge.	Accuracy, Completeness, Consistency, Timeliness, Interpretability, Security, Trustworthiness.	Accuracy: This measures whether the data reflects real-world facts correctly; it is the degree to which data is close to the realistic value. In practical metrics, it is often defined as the percentage of correct facts or triples within the knowledge graph. Completeness: This describes whether the dataset contains all relevant data of interest. It relates to the "coverage" of the dataset and the problem of whether the graph contains all necessary facts. Consistency: This means that the data agrees with itself and is free of conflicts regarding particular integrity constraints. It often involves identifying assertions that violate logic rules or ontology axioms. Timeliness: This reflects the degree to which the data is up-to-date. It is particularly critical for dynamic datasets that change frequently. Trustworthiness: While a single-sentence definition is not explicitly provided, the sources link it to the credibility of the entries. It is measured by the degree to which a triple is likely to be correct, often using models that fuse entity, relationship, and global graph features to output a "trustworthy" value. Security: The sources categorize security as a dimension of Accessibility, which refers to the ease with which data can be retrieved and the protections surrounding that access. (Note: A more detailed technical definition for "Security" is not provided in the excerpts). Interpretability: This dimension describes how easily the results or methods can be understood by humans. For instance, rule-based methods are noted for having high interpretability because they represent knowledge explicitly, whereas embedding-based deep learning methods are described as having poor interpretability due to their complex parameter structures.	<i>B. Xue and L. Zou, "Knowledge Graph Quality Management: A Comprehensive Survey," IEEE Transactions on Knowledge and Data Engineering (TKDE), 2022, doi: 10.1109/TKDE.2022.3150080. Read</i>
23	Knowledge Graph Quality Control: A Survey	Xiangyu Wang, Lyuzhou Chen, Taiyu Ban, Muhammad Usman, Yifeng Guan, Shikang Liu, Tianhao Wu, Huanhuan Chen.	2021	Surveys Linked Data/KG quality control. Defines six evaluation dimensions (e.g., completeness, accuracy, timeliness, trust) and analyzes their relationships. Reviews how to identify and mitigate errors during KG construction and maintenance.	Provide a comprehensive overview of KG quality control dimensions and corresponding techniques.	Literature survey of KG quality dimensions; classification of control/treatment techniques (e.g., data cleaning, constraint enforcement) across the KG lifecycle.		Survey (no single model)		Clearly defines multiple quality dimensions; discusses both theoretical and practical treatments of quality issues.	Survey only; may not cover very latest work; focuses on Linked Data (RDF) context.	Accuracy, Completeness, Timeliness, Trustworthiness (Believability), Consistency, Security.	Accuracy: This refers to the degree of correctness of facts within a KG. It is also defined as the extent to which knowledge is "correct, reliable, and certified free of error". It focuses on whether the data values correctly represent real-world states. Completeness: This is defined as the extent to which a KG covers the knowledge of interest. Because "knowledge of interest" depends on the specific use case, completeness is considered a relative dimension; a KG is complete if it meets the requirements of its underlying applications. • Timeliness: This dimension refers to the degree to which knowledge is up-to-date. It can be measured by the time range since the last update or verification of a statement. Trustworthiness (Believability): This is the degree of objectivity, authority, and verifiability of a KG. The sources describe it as how credible the information is accepted to be (i.e., whether it is accepted as real, true, and correct). Consistency: This is defined as the degree to which the knowledge in a KG does not contradict itself. Unlike accuracy, which focuses on semantic correctness (truth), consistency focuses on logical correctness and ensuring the knowledge base is free of formal contradictions. Security: While the paper mentions this dimension in a table of related frameworks, the authors categorize it as irrelevant or redundant to the core evaluation of KGs. It is defined in the sources as the "extent to which information is protected from harm in the context of a particular activity".	<i>X. Wang et al., "Knowledge graph quality control: A survey," Fundamental Research, vol. 1, no. 5, pp. 607-626, 2021, doi: 10.1016/j.fmre.2021.09.003. Read</i>
24	Knowledge Graph Refinement: A Survey of	Heiko Paulheim.	2017	Surveys <i>KG refinement</i> (completeness and	Provide an overview of techniques and evaluation	Literature survey of KG completion/correction;		Survey (no single model)		Comprehensive review of KG improvement methods	Focus on methods (not metrics per se); older	Accuracy, Completeness, Consistent	Accuracy (Correctness / Free-of-Error): The sources relate this dimension to the correctness of a knowledge graph. Specifically, this is linked to the data quality dimension of being "free-of-error," which is one of the primary goals of knowledge graph refinement. Refinement for accuracy involves identifying and	<i>H. Paulheim, "Knowledge graph refinement: A survey of</i>

	Approaches and Evaluation Methods.			error-correction). Reviews methods for inferring missing facts and detecting errors in KGs, along with their evaluation. Summarizes both completion and correction techniques, and categorizes evaluation methodologies.	practices for improving KG quality.	categorizes methods (rule-based, ML, inference) and evaluation strategies (gold vs. silver standard, etc.)				(completion and cleaning) with attention to how they are evaluated.	(2016), so missing newer embedding/LM approaches.	representation, Refinement (Value added)	removing erroneous pieces of information to increase the overall reliability of the data. Completeness (Coverage): This dimension is referred to as coverage or completion. The author defines full coverage as a state where a knowledge graph contains "information about each and every entity in the universe," though he notes this is not reasonably reachable. In practice, completeness refinement focuses on addressing "the shortcoming of incompleteness" by adding missing knowledge to the graph. There is typically a noted trade-off between coverage and correctness. Consistent Representation (Consistency): While the paper does not use the specific phrase "consistent representation" as a formal definition, it focuses heavily on consistency through logical reasoning. This involves "validating the consistency of statements" to ensure they do not contradict the schema or ontology. For example, a statement is inconsistent if it assigns an entity to two "disjoint" classes, such as a single entity being both a person and a city. Refinement (Value Added): Refinement is defined as the process of improving an existing knowledge graph, as opposed to "construction," which is building one from scratch. The "value added" or utility of refinement comes from its ability to increase the graph's effectiveness and impact. By decoupling refinement from construction, researchers can develop generic methods that improve multiple graphs and allow for a better understanding of the approach's actual effectiveness.	<i>approaches and evaluation methods," Semantic Web, vol. 8, pp. 489-508, 2017, doi: 10.3233/SW-160218. Read</i>
25	Linked Data Quality Assessment through Network Analysis	Christophe Guéret, Paul Groth, Claus Stadler, Jens Lehmann.	2011	Proposes a framework to assess Linked Data quality from a global network perspective. Uses graph/network measures (e.g., degree, connectivity) to quantify whether newly added inter-dataset links improve or degrade data quality. Validated on known-quality link sets to detect quality changes.	Determine at a global scale if adding links improves or hurts Linked Data quality using network properties.	Builds global Linked Data graph (or subset), computes statistical network metrics (e.g., clustering, centralization) before and after adding links. Changes in metrics indicate quality improvement or degradation.	LINK-QA (network-statistics-based QA).	Network analysis framework.	Validated on test cases of known good/bad links (effective at detecting quality changes).	Does not require manual gold data; provides global summary of link quality; extensible by adding more network metrics.	Focuses only on connectivity/structure (not semantics or content); may miss subtle data issues; requires selecting appropriate metrics.	Degree, Clustering Coefficient, Open owl:sameAs Chains, Centrality, Description Richness.	Degree: This is defined as the number of connections (links) associated with a resource. The paper assesses quality by checking if the addition of links helps the network move toward a power-law distribution of the degree. Clustering Coefficient: This dimension analyzes the extent to which nodes in the graph tend to cluster together. It is one of the statistical techniques used to provide a summary of the network's global properties. Open owl:sameAs Chains: This is a measure that counts the number of open identity chains within a local network. It is used to assess how new links impact the structural identity and connectivity of the Web of Data. Centrality: This dimension measures the relative importance or structural prominence of a resource within the network. Like the other metrics, it is aggregated into a distribution to provide an overall perspective on the network's quality. Description Richness: This is defined as a measure of the amount of descriptive information available for a resource. A primary goal of adding high-quality links, according to the framework, is to increase the overall descriptive richness of the entities involved.	<i>C. Guéret, P. Groth, C. Stadler, and J. Lehmann, "Linked Data Quality Assessment through Network Analysis," in Proc. ISWC Posters & Demos, 2011. Read</i>
26	Harnessing Diverse Perspectives: A Multi-Agent Framework for Enhanced Error Detection in	Yu Li, Yi Huang, Guilin Qi, Junlan Feng, Nan Hu, Songlin Zhai, Haohan Xue, Yongrui Chen,	2025	Introduces MAKGED, a multi-agent LLM+GCN approach for KG error detection.	Enhance triple error detection by combining structured graph views with LLM reasoning and	Build bidirectional subgraphs for head/tail of each triple; use GCN to embed structure and	MAKGED (LLM+GCN multi-agent).	Multi-agent (LLM + GCN).	Achieves 0.73% (FB15k) and 6.62% (WN18RR) higher accuracy	Integrates fine-grained graph info with LLM knowledge; multi-agent discussion yields	High computational and implementation complexity; requires multiple	Accuracy, Objectivity (Consensus), Interpretability, Reliability, Consistent	Accuracy: This is the primary metric used to evaluate the framework's performance in identifying whether a knowledge graph triple is correct or incorrect. The paper defines the goal of MAKGED as improving error detection accuracy by integrating structural and semantic representations, outperforming state-of-the-art methods on standard datasets like FB15K and WN18RR. Objectivity (Consensus): While "Objectivity" is not a formal term used, the paper defines Consensus as the state achieved when all agents agree on the correctness	<i>Y. Li et al., "Harnessing Diverse Perspectives: A Multi-Agent Framework for Enhanced Error</i>

	Knowledge Graphs (MAKGED).	Ruoyan Shen, Tongtong Wu.		Four agents (head-forward/backward, tail-forward/backward) are trained on bidirectional subgraph embeddings concatenated with LLM query embeddings. They discuss and vote on each triple's correctness. Outperforms SOTA by +0.73% on FB15k and +6.62% on WN18RR.	multi-agent consensus.	Llama2 for semantic embedding; train four specialized agents; agents independently evaluate triples then engage in multi-round discussion and vote on final decision.			than previous best methods.	transparent, robust decisions; strong gains on benchmarks.	LLMs/agents; performance depends on quality of subgraphs and LLM reasoning; novel (practical impact TBD).	representation.	of a triple during a consistency check. To ensure a fair and objective outcome when agents disagree, the framework employs a majority rule strategy after three rounds of discussion, or utilizes a "summarizer agent" to break a 2-vs-2 tie by reflecting on the collective reasoning. Interpretability: The sources refer to this as Transparency and Clarity. It is defined as a decision-making process that avoids the "black box" approach of existing methods—which typically only provide a single confidence score—by allowing agents to engage in collaborative evaluation and reasoning. This provides clear "context about how decisions are made". Reliability: The paper defines the need for error detection as a "crucial" step for "ensuring the reliability of downstream applications," such as recommendation systems and retrieval-augmented generation. By using bidirectional subgraph embeddings, the framework specifically aims to enhance the robustness of these evaluations in both general and industrial scenarios. Consistent Representation: The paper describes this as a "unified embedding representation". It is defined as the result of concatenating structural features (fine-grained, bidirectional subgraph embeddings generated by a GCN) with semantic features (text-based query embeddings from an LLM). This creates a richer, more expressive input that allows agents to evaluate triples from multiple perspectives simultaneously.	<i>Detection in Knowledge Graphs," 2025, arXiv: 2501.15791.</i> Read
27	Linked Data – Design Issues.	Tim Berners-Lee.	2006	The document outlines the fundamental design principles behind Linked Data on the Web. It proposes a set of rules for how data should be identified, accessed, described, and linked to other resources to enable a global web of data. These ideas became the cornerstone of the Semantic Web and inspired	To define the conceptual and technical foundations of Linked Data; to guide how web resources can be structured and connected using URIs and RDF so that both humans and machines can discover and interpret data efficiently.	Conceptual design proposal based on Web architecture principles. Rather than an empirical or experimental study, Berners-Lee formulates prescriptive <i>design rules</i> derived from early web development experience and Semantic Web goals.	No computational algorithm. The core “model” is a set of four Linked Data principles: (1) Use URIs as names for things, (2) Use HTTP URIs, (3) Provide useful RDF information upon lookup, (4) Include links to other URIs for discovery.	The Linked Data Principles and, indirectly, the 5-Star Open Data model that extends these rules.	Not a quantitative or empirical work; accuracy is conceptual. The “validation” comes from global adoption and the successful growth of the Linked Open Data cloud following these principles.	- Established the foundation of Linked Data and the Semantic Web. - Introduced reusable, web-scalable identification and linking mechanisms. - Influenced all later data quality, interoperability, and accessibility frameworks (FAIR, DQV, ISO 25012, Zaveri 2015, etc.). - Promoted openness and machine-readability of web data.	- No formal metrics or measurable criteria for quality. - Lacks guidance on assessing data correctness, completeness, or trust. - Doesn't consider governance, provenance, or evolving knowledge graph structures. - Functions as a conceptual manifesto, not a	Accessibility, Interpretability, Relevancy (Interlinking), Consistent representation, Interoperability.	Accessibility: It is defined as the act of making information publicly available on the web rather than having it "buried in a zip archive somewhere". It involves using HTTP URIs as names for things so that people and machines can "look up those names" and retrieve information. In the 5-star rating system, the first level of accessibility is achieved simply by making the data available on the web in any format under an open license. Interpretability: It refers to the requirement that data be machine-readable and structured so that it can be processed by a computer rather than just viewed by a human. To be interpretable, data should avoid flat image scans and instead use structured formats (like Excel or CSV) and eventually standardized frameworks like RDF, which allow a "person or machine" to explore and understand the relationships between arbitrary things. Relevancy (Interlinking): It is the practice of including links to other URIs within your data to connect it to the wider web. This dimension is considered the "value added by the web" because it allows for the unexpected re-use of information and provides context by connecting local data to related external material. Without these links, data remains an isolated silo rather than part of a "serious, unbounded web". Consistent Representation: It involves using a universal set of symbols (URIs) to identify objects or concepts so that anyone on the planet can refer to them globally. The sources note that consistency can be challenging when data is repeated across different documents to make it "browsable," as this requires coordination between different authors or programs to ensure the information remains the same across the web. Interoperability: It is the ability of different systems to "point at your stuff" and integrate it into a single web of data. This is achieved by moving from proprietary formats to non-proprietary formats (like CSV) and utilizing W3C open standards such as RDF and SPARQL. It allows data from disparate	<i>T. Berners-Lee, "Linked Data - Design Issues," Jul. 27, 2006 (Updated 2009).</i> Read

				many later frameworks for data and knowledge graph quality.							testable framework.		sources—such as different social network portals—to be "linked together into one web" because they expose their data in a standard form.	
28	F-UJI and FAIR-Checker: Automated FAIR Data Assessment Tools	Robert Huber, and Anusuriya Devaraju.	2020	F-UJI and FAIR-Checker are automated tools that operationalize FAIR principles for data and metadata evaluation. They test digital resources against Findable, Accessible, Interoperable, and Reusable criteria and provide compliance scoring and actionable improvement recommendations.	Enable reproducible, objective, and automated FAIRness assessment for datasets and knowledge graphs.	Rule-based and API-driven automated FAIR metric evaluation. Includes data harvesting, metadata inspection, and API validation.	FAIR-Checker API; F-UJI Web Service	FAIR Assessment Tools	Quantitative FAIRness scoring and compliance benchmarking	Automates FAIR compliance checks; improves interoperability and transparency; provides concrete, machine-actionable feedback for repository improvement.	Limited to FAIR dimensions; may not capture semantic richness or structural ontology quality directly.	Findability, Accessibility, Interoperability, Reusability.	• Findability: This dimension ensures that both humans and computer systems can easily locate data and metadata. It usually requires assigning a persistent identifier (PID) and providing rich metadata that is indexed in a searchable resource. • Accessibility: Once data is found, users need to know how it can be accessed. This dimension specifies the protocols for data retrieval, including whether authentication or authorization is required. • Interoperability: This refers to the ability of data to be integrated with other datasets and to function across different applications or workflows. It requires the use of standardized vocabularies and formats. • Reusability: The goal is to ensure data can be used for future research. This requires clear usage licenses and detailed provenance (history) so that others understand the context and limitations of the data.	<i>A. Devaraju and R. Huber, "F-UJI - An Automated FAIR Data Assessment Tool," Zenodo, Oct. 02, 2020, doi: 10.5281/zenodo.4063720. Read</i>
29	ISO/IEC 25024 & ISO 8000: Data Quality Measurement and Metadata Standards	Technical Committee ISO/TC 184, Automation systems and integration, Subcommittee SC 4, Industrial data.	2015	ISO/IEC 25024 extends ISO 25012 by defining formal methods for measuring data quality, while ISO 8000 introduces standardize	Define measurable and auditable data quality and metadata quality metrics beyond conceptual modeling.	International standardization process defining attributes, measures, and compliance verification for data quality evaluation.	ISO/IEC 25024 Measurement Model; ISO 8000 MDM Framework	Data Quality Measurement Standards	Used in enterprise-level DQ audits; compliance benchmarked via quantifiable characteristics	Provide measurable, auditable quality control; complement KG governance frameworks with enterprise-grade validation.	Not graph-specific; limited integration with semantic web and RDF-based KGs without mapping extensions.	Accuracy, Completeness, Consistency, Timeliness, Security, Compliance, Traceability.	Accuracy: The degree to which data correctly represents the real-world entities or values they are intended to model. Completeness: The degree to which all required data is present and no essential information is missing. Consistency: The degree to which data is uniform and coherent across different datasets, systems, or time periods. Timeliness: The degree to which data is up-to-date and available when needed for decision-making or processing. Security: While the sources do not provide a single definition for "Security," they address it through. ◦ Confidentiality is the degree to which data is protected from unauthorized access and disclosure. ◦ Integrity is the degree to which data is protected from unauthorized modification and maintains its accuracy over time.	<i>Systems and software engineering — Measurement of data quality, ISO/IEC 25024:2015, 2015. Read</i>

				d data governance and master data management guidelines.									Compliance: The degree to which data conforms to applicable regulations, standards, and organizational policies. Traceability: The degree to which the origin, processing history, and lineage of data can be tracked and verified.	
30	The FAIR Guiding Principles for scientific data management and stewardship	Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble,	2016	Proposed guiding principles for research data stewardship (Findable, Accessible, Interoperable, Reusable).	To provide universally applicable guidance for data stewardship and optimize reuse.	Conceptual Framework Development.	N/A (Principles)	FAIR Principles	Global Standard (Widely Adopted)	Defines essential governance/metadata requirements; pathway for systematic sharing.	Requires addressing significant ethical and technical challenges for full compliance.	Findability, Accessibility, Interoperability, Reusability.	Findability: Findability ensures that data and metadata are easy to discover by both human researchers and computational agents. To be findable: • (Meta)data must be assigned a globally unique and persistent identifier. • Data must be described with rich metadata. • Metadata must explicitly include the unique identifier of the data it describes. • (Meta)data must be registered or indexed in a searchable resource. Accessibility: Accessibility focuses on the ability of users to reach the data once it has been found. To be accessible: • (Meta)data must be retrievable by their identifier using a standardized communications protocol. • The protocol used must be open, free, and universally implementable, allowing for authentication and authorization where necessary. • Metadata must remain accessible even if the data itself is no longer available. Interoperability: The sources formally define interoperability as "the ability of data or tools from non-cooperating resources to integrate or work together with minimal effort". To achieve this: • (Meta)data must use a formal, accessible, shared, and broadly applicable language for knowledge representation. • (Meta)data must use vocabularies that also follow the FAIR principles. • (Meta)data must include qualified references to other (meta)data. Reusability: Reusability is the ultimate goal of the FAIR principles, aiming to ensure that data can be effectively integrated and used in future investigations. To be reusable: • (Meta)data must be richly described with a plurality of accurate and relevant attributes. • (Meta)data must be released with a clear and accessible data usage license. • (Meta)data must be associated with detailed provenance. • (Meta)data must meet domain-relevant community standards.	<i>M. D. Wilkinson et al., "The FAIR Guiding Principles for scientific data management and stewardship," Scientific Data, vol. 3, 2016, Art. no. 160018, doi: 10.1038/sdata.2016.18.</i> Read

[illegible]

		and Barend Mons												
31	How to Implement a Knowledge Graph Completeness Assessment with the Guidance of User Requirements	ZHANG Ying and XIAO Gang	2024	Addresses completeness dimension; clarifies concepts and metrics aligned with user needs.	Develop user-guided metrics/measurement proposal for completeness.	Theoretical Framework.	N/A (Conceptual)	User-Guided Completeness Framework	N/A (Conceptual).	Aligns completeness with user context; provides clear definitions.	Focus solely on completeness; framework-level only.	Completeness, Value added, Appropriate amount (Coverage)	Completeness: The sources distinguish between a traditional definition and the authors' newly proposed concept: • Traditional Completeness: Generally refers to the "ratio of the current data volume to the total data volume". • Relative Completeness: This is the paper's core definition, described as the "ratio of the amount of knowledge contained in the knowledge graph to the amount of knowledge required by the user in a specific context of use at a specific time". It emphasizes that because knowledge is infinite and constantly updating, completeness must be measured within a "reasonably limited range" defined by actual user needs. To calculate this, the paper further defines three sub-metrics: • Entity Completeness: The "ratio of the number of entities in the knowledge graph to the number of entities that meet the user's query requirements". • Schema Completeness: The "proportion of classes or properties contained in the knowledge graph that satisfy the user's query requirements". • Property Completeness: The "ratio of the number of property values of a particular property... in the knowledge graph to meet the number of property values demanded by users' queries". Appropriate Amount (Coverage) The paper addresses "Appropriate Amount" under the definition of the "amount of knowledge" and its "coverage": • Amount of Knowledge: This refers to the "breadth of knowledge covered by the knowledge graph". It is measured quantitatively by the number of triples, entities, relationships, or related properties that can be retrieved. • Data Coverage: The authors note that when choosing a graph for a specific task, users must consider "data coverage" and the "amount of data" to ensure the graph contains the specific information needed for the current objective. They emphasize that a large scale or volume does not necessarily mean the graph is complete for a specific user's needs. Value Added The term "Value Added" is not defined as a formal quality dimension in the paper. However, the sources describe the "value" and utility provided by their proposed method in the following ways: • Reference Value: The method provides a "better reference value" for users compared to traditional assessments that lack requirement awareness. • Applicability: By mining user trends and latent needs, the method provides an "informed reference" that enhances the overall "applicability of the knowledge graph". • Optimization Value: The assessment identifies missing knowledge, which helps builders "optimize and improve the construction and updating process" of the graph in a targeted, time-saving, and efficient manner.	Y. Zhang and G. Xiao, "How to implement a knowledge graph completeness assessment with the guidance of user requirements," <i>Journal of Systems Engineering and Electronics</i> , vol. 35, no. 3, pp. 679-688, 2024. Read

32	Knowledge Graph Completeness : A Systematic Literature Review	Subhi Issa, Onaopepo Adekunle, Fayçal Hamdi, Samira Si-Said Cherfi, Michel Dumontier, and Amrapali Zaveri.	2021	Reviews KG completeness research. Defines completeness types and surveys metrics.	Synthesize how completeness is defined and measured; identify trends and gaps.	Systematic Review (ISO guidelines).	N/A (Survey)	N/A	N/A	Thorough coverage of the completeness dimension; clarifies taxonomy of issues.	Focused only on completeness; does not propose new methods.	Completeness , Coverage, Relevancy (Missing Data), Appropriate Amount.	Completeness: It is a data quality measure that refers to the amount of information present in a particular dataset. It identifies and defines seven specific dimensions (or types) of completeness, which are: Schema completeness: It is the degree to which the classes and properties of an ontology are represented in a Linked Data dataset. Property completeness: It is the degree to which values for a specific property are available for a given task. Population completeness: It is the degree to which all real-world objects of a particular type are represented in a Linked Data dataset. Interlinking completeness: It is the degree to which instances are interlinked in a Linked Data dataset with respect to some reference dataset(s). Currency completeness: It is the degree to which elements of a knowledge base are available as it is updated over time. Metadata completeness: It is the degree to which metadata properties and values are not missing in a dataset for a given task. Labelling completeness: It refers to the degree to which entities in the dataset have a human readable label.	<i>S. Issa, O. Adekunle, F. Hamdi, S. S. -S. Cherfi, M. Dumontier and A. Zaveri, "Knowledge Graph Completeness: A Systematic Literature Review," in IEEE Access, vol. 9, pp. 31322-31339, 2021, doi: 10.1109/ACCESS.2021.3056622. Read</i>
33	Towards assessing the quality of knowledge graphs via differential testing	Jiajun Tan, Dong Wang, Jingyu Sun, Zixi Liu, Xiaoruo Li, Yang Feng.	2024	Proposes automated differential -testing: train multiple KG embedding models and compare outputs.	Automaticall y quantify overall KG quality (correctness) without gold data.	Differential Testing / Ensemble Learning.	Differential-KGEM (Ensemble)	Differential-KGEM	High Correlation (Distinguishes error levels).	Requires no external ground truth; leverages consensus to detect noise.	Dependent on choice/number of models; sensitive to hyperparameters.	Accuracy, Consistent representation, Reliability, Objectivity (Consensus).	Accuracy: Traditionally, KG quality is measured by the accuracy of test data. The authors aim for their automated approach to be capable of "accurately evaluating" the quality of KGs compared to manual methods. Reliability: The paper states that its approach produces reliable results across different datasets. It also notes that manual annotations are considered a "reliable" but high-cost gold standard. Consistency (Consistent Representation): The paper evaluates quality by comparing the consistency of predictions from different Knowledge Graph Embedding Models (KGEMs). If the results from multiple models are similar (consistent), the quality of the KG is considered higher. Objectivity (Consensus): The methodology is based on the "majority decision" principle of differential testing, which assumes that if most models agree on a result, it is likely correct. This consensus serves as a cross-referencing "oracle" to address the lack of standard manual answers.	<i>Tan, Jiajun & Wang, Dong & Sun, Jingyu & Liu, Zixi & Li, Xiaoruo & Feng, Yang. (2024). Towards assessing the quality of knowledge graphs via differential testing. Information and Software Technology. 174. 107521. 10.1016/j.infsof.2024.107521. Read</i>
34	Triple Trustworthiness Measurement for Knowledge Graph.	Shengbin Jia, Yang Xiang, and Xiaojun Chen.	2019	Neural model assessing trustworthiness of triples by fusing local/global features.	Quantify how likely each triple is correct using semantics and context.	Neural Network (Multi-view Fusion).	KGTrm (Criss-cross Neural Network)	KGTrm	Acc: ~0.977 (FB15k).	Captures multiple evidence sources (entity semantics, relation patterns).	Complex model requiring training; assumes access to KG structure; low interpretability.	Accuracy, Objectivity, Trustworthiness.	Trustworthiness: The paper introduces the concept of KG triple trustworthiness to indicate the "degree of certainty that the knowledge expressed by the triple is true". It is represented as a value within the interval [0, 1], where a value closer to 0 indicates a greater probability that the triple contains an error. Accuracy: While the specific term "Accuracy" is used as a performance metric for the model's experimental results , the paper uses the term "semantic correctness" to describe the qualitative goal of their measurement, which is to quantify the "true degree of the facts expressed" in a triple. Objectivity: It describes the Knowledge Graph as a means to "describe the various entities or concepts and their relationships existing in the objective world". The authors aim to measure trustworthiness by evaluating how well a triple integrates into the "social group" of the KG, assuming that only true triples can achieve popular recognition from other relevant triples.	<i>Shengbin Jia, Yang Xiang, Xiaojun Chen, Kun Wang, and Shijia. 2019. Triple Trustworthiness Measurement for Knowledge Graph. In The World Wide Web Conference (WWW '19). Association for Computing Machinery, New York, NY, USA, 2865–2871. Read</i>

35	Error Detection on Knowledge Graphs with Triple Embedding	Yezi Liu, Qinggang Zhang, Mengnan Du, Xiao Huang, and Xia Hu.	2023	Unsupervised framework treating triples as nodes in a triple network (BiLSTM+GAT).	Detect erroneous triples without labeled data leveraging patterns/context.	Graph Attention Network (GAT) + BiLSTM.	TripleNet (BiLSTM + GAT)	TripleNet	High Precision /F1 (Outperforms SOTA).	Unsupervised (no labels needed); captures sequential and structural info.	Complex; performance depends on network size/sparsity.	Accuracy, Consistent representation, Reliability, Structural Consistency	Accuracy: Defined as the precision and correctness of the triples within a knowledge graph. The paper notes that even high-quality graphs like YAGO3 have an accuracy of 95%, while others like NELL have an estimated precision of only 74%. Consistent Representation (Inter-triple Context): While not used as a single term, the paper discusses capturing the "contextual information" and "inter-dependencies among triples" to ensure that a triple is semantically consistent with its neighbors. Reliability (Suspicious Score): The paper uses a "suspicious score" to measure the likelihood of a triple being an error. A higher suspicious score indicates that a triple is less reliable and more likely to be a "noisy fact". Structural Consistency (Translational Structure): This refers to the "translational or sequential structure inside a triple". The model uses a Bi-LSTM to ensure that the head, relation, and tail components follow the expected structural patterns learned from the graph.	<i>Y. Liu, Q. Zhang, M. Du, X. Huang and X. Hu, "Error Detection on Knowledge Graphs with Triple Embedding," 2023 31st European Signal Processing Conference (EUSIPCO), Helsinki, Finland, 2023, pp. 1604-1608, doi: 10.23919/EUSIPCO58844.2023.10289852.</i> Read
36	Contrastive Knowledge Graph Error Detection.	Qinggang Zhang, Junnan Dong, Keyu Duan, Xiao Huang, Yezi Liu, Linchuan Xu.	2022	Unsupervised detector using contrastive learning; views triples as nodes in hyper-graphs.	Detects KG errors modeling multi-view consistency without rules.	Contrastive Learning (Multi-view).	CAGED (Contrastive Multi-view)	CAGED Model	Significantly better than SOTA.	No manual negatives needed; leverages multi-view consistency for subtle errors.	Computational overhead of contrastive training; needs clear view partitioning.	Accuracy, Consistent representation, Reliability, Internal Coherence.	Accuracy: The paper defines an optimal rating model p_{θ} that should accurately denote the probability of a triple (h,r,t) being a defined error. Performance is measured through Precision and Recall of the triples identified with the lowest confidence scores. Consistent Representation (Consistency of triple representations across multi-views): This refers to measuring how well a triple's representations, learned from different augmented "hyper-views" (View I and View II), can be pulled close to each other. If a triple is normal, its representations across these views should be consistent. Reliability (Reliable KG embeddings): The paper aims to produce reliable KG embeddings by using an error-aware graph encoder (EaGNN). This encoder is designed to block information propagation from potential errors, ensuring the learned embeddings are robust and dependable for error detection. Internal Coherence (Self-consistency within the triple): This is the degree to which a triple fits the translation assumption (i.e., $h+r=t$). It reflects the self-contradictory nature (or lack thereof) within the triple itself, where higher unconformity indicates a likely error.	<i>Qinggang Zhang, Junnan Dong, Keyu Duan, Xiao Huang, Yezi Liu, and Linchuan Xu. 2022. Contrastive Knowledge Graph Error Detection. In Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM '22).</i> Read